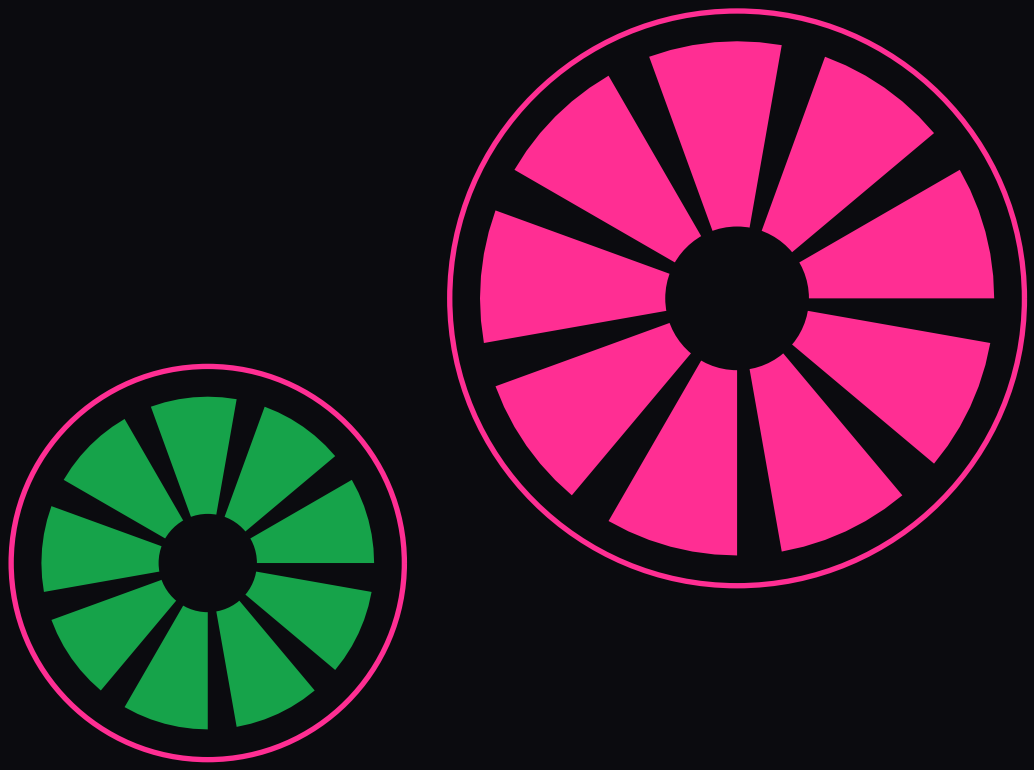


COMPUTER HARDWARE | ASSIGNMENT

GPU & Cooling Systems

GPU identification - Cooling components - Air vs liquid - Gaming build

TASK 1 Identify the GPU Installed in Your System

A **GPU (Graphics Processing Unit)** renders images, video and games. An **integrated** GPU is built into the CPU and shares system RAM; a **dedicated (discrete)** GPU is a separate card or chip with its own video memory (VRAM). Many laptops have both.

How to find it

1	**Windows (quickest):** press Ctrl + Shift + Esc, open **Performance** and select **GPU**. The model name is shown at the top right. If you see GPU 0 and GPU 1, one is integrated and the other dedicated.
2	**DirectX tool:** press Win + R, type **dxdiag**, and open the **Display** tab.
3	**Device Manager:** right-click Start, open Device Manager and expand **Display adapters**.
4	**Linux / macOS:** run **lspci | grep -i vga** on Linux, or open **About This Mac** on macOS.
5	**No computer?** Use an online PC-building/hardware simulator and read the GPU model from its parts list.

Quick guide: integrated vs dedicated

	Integrated	Dedicated
Examples	Intel UHD / Iris Xe, AMD Radeon Graphics (in Ryzen APUs)	NVIDIA GeForce RTX / GTX, AMD Radeon RX, Intel Arc
Memory	Shares system RAM	Has its own VRAM
Best for	Browsing, office work, video streaming, light games	Gaming, video editing, 3D work, AI tasks

My GPU details (fill in)

GPU brand (NVIDIA / AMD / Intel)	
Model number	
Integrated or dedicated?	
VRAM (if dedicated)	
Method used to find it	

PASTE YOUR PHOTO / SCREENSHOT HERE

Screenshot of Task Manager / dxdiag / simulator showing the GPU

TASK 2 — Photo of the Cooling System & Component Labels

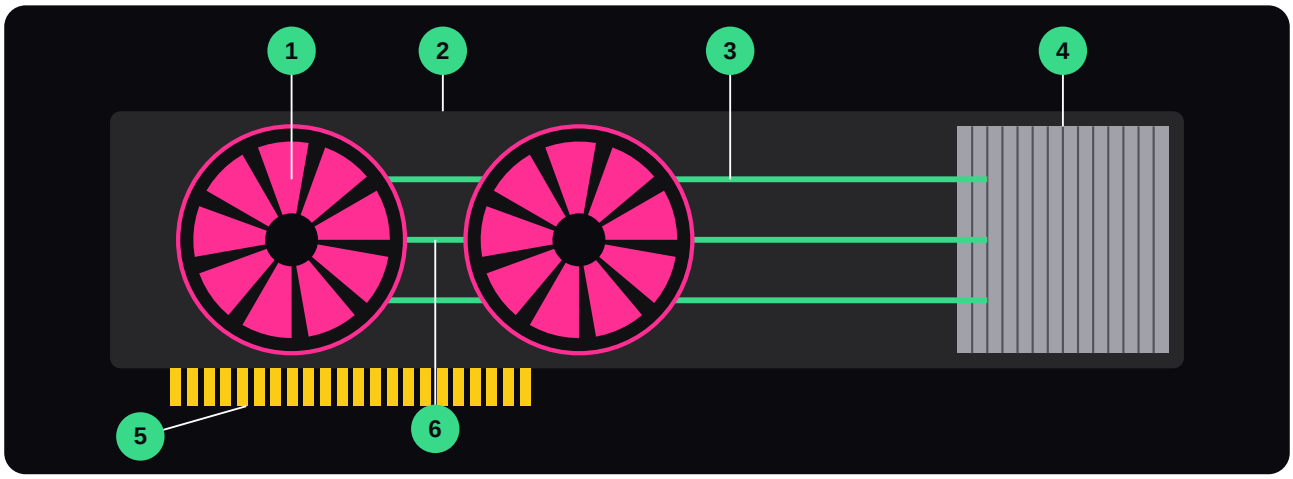
Safety: Switch off and unplug the PC before opening the case, and touch bare metal to discharge static. Never touch fan blades while powered. If you cannot open a PC, take a screenshot from a hardware simulator or use a clear product photo you have permission to use.

Your photo (label it with the numbers below)

PASTE YOUR PHOTO / SCREENSHOT HERE

Fan, heatsink or liquid cooler inside the PC / simulator

Reference diagram: parts of an air-cooled graphics card



#	Component	Function
1	**Fan**	Pushes air across the heatsink fins so hot air is carried away from the GPU.
2	**Shroud**	Plastic/metal cover that guides airflow and protects the cooler.
3	**Heat pipes**	Sealed copper tubes that quickly move heat from the GPU chip to the fins.
4	**Heatsink (fins)**	Metal fins with a large surface area that absorb heat and release it into the air.
5	**PCIe connector**	Gold contacts that plug into the motherboard slot for data and power.
6	**GPU die + thermal paste (under cooler)**	The chip that produces heat; thermal paste fills tiny gaps so heat transfers to the cooler plate.
+	**Liquid cooling parts (if present)**	**Water block** sits on the GPU, **pump** circulates coolant, **tubing** carries it, and the **radiator + fans** release the heat.

TASK 3

Air Cooling vs Liquid Cooling for GPUs

Most graphics cards ship with built-in air coolers. Liquid cooling uses coolant, a pump and a radiator to move heat away from the GPU. Their main differences are compared below.

Factor	Air cooling	Liquid cooling
Cost	Included with the card; no extra spending. Upgraded aftermarket coolers are fairly cheap.	Noticeably more expensive: liquid-cooled cards, AIO kits or custom loops cost extra, and custom loops cost the most.
Cooling efficiency	Good for most gaming. Can run hotter and louder under long heavy loads, especially in a hot room.	Usually keeps temperatures lower and steadier under heavy loads, with room for overclocking.
Maintenance	Very low: clean dust from fans and fins, and replace thermal paste occasionally.	Higher: check pump, tubes and radiator; custom loops need coolant changes. Sealed AIO units need less work.
Noise	Fans can get loud at high load.	Often quieter because large radiators run slower fans, though the pump adds a faint hum.
Installation & space	Simple: it is part of the card, nothing extra to mount.	Complex: needs a radiator mount in the case, careful tube routing and compatibility checks.
Reliability / risk	Few parts; a failed fan is cheap to replace.	More parts that can fail (pump, leaks); a leak could damage components, though rare in sealed units.

Summary: Air cooling wins on cost, simplicity and reliability. Liquid cooling wins on temperatures, noise control and extreme performance, at a higher price and with more upkeep.

TASK 4 My Gaming + IPL Streaming PC: Cooling Choice

Scenario: A PC for streaming IPL matches and playing graphics-heavy games.

My choice: AIR COOLING for the GPU.

Justification

I would choose air cooling because modern dedicated GPUs with two or three large fans handle graphics-heavy games well, and streaming IPL matches is a light task that barely stresses the GPU. Air cooling costs nothing extra, so that budget can go into a stronger GPU or more RAM, which improves gaming more than a fancier cooler would. It also needs only occasional dust cleaning, which suits Indian conditions where dust is common, and it avoids the pump failure and leak risks of liquid systems. Liquid cooling would make sense only for a heavily overclocked, very high-end card, so for a normal gaming and streaming build a well-ventilated case with good air cooling is the practical choice.

Supporting points

Need	Why air cooling fits
Graphics-heavy games	Triple-fan air coolers keep most mid to high-end GPUs within safe temperatures.
IPL streaming	Video playback and streaming use little GPU power, so heat output stays low.
Budget	No extra cost; savings can go towards a better GPU.
Upkeep	Only dust cleaning and occasional thermal paste replacement.
Hot, dusty climate	Good case airflow plus dust filters work well; no coolant or pump to monitor.

Conclusion: Knowing your GPU, how it is cooled and which cooling method fits your needs helps you build a faster, quieter and longer-lasting PC.